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The Measurement, Control and Targeting of Housing Finance Subsidies: The Case of Argentina

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Discussion Paper

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ABSTRACT

This paper examines the kinds of housing subsidies that are channeled through housing finance systems in economies experiencing high inflation and financial stress. While Argentina is used as a case study, the kinds of programs analyzed appear to be very similar to those in operation in other economies.

The paper presents measures of the subsidy level and welfare costs of housing policy. It shows that both the subsidies and the welfare costs have been very large. Policies that would improve the targeting and transparency of the subsidies, and reduce the welfare losses are discussed.

THE MEASUREMENT, CONTROL AND TARGETING OF HOUSING FINANCE SUBSIDIES: THE CASE OF ARGENTINA

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THE MEASURMENT, CONTROL AND TARGETING OF HOUSING FINANCE SUBSIDIES: THE CASE OF ARGENTINA

I. INTRODUCTION

This paper examines the kinds of housing subsidies that are often channeled through housing finance systems in economies experiencing high inflation and financial stress. Argentina is used as a case study. However, the kinds of programs analyzed appear to be very similar to the programs in operation in a number of other economies.[1/] The purpose of the paper is threefold.

First, to present simple measurements of both the subsidy level and welfare costs of housing policy. A user-cost of capital approach is used to measure the subsidy level. The main advantage of this approach is that it shows how sensitive the subsidy levels are to changes in economic conditions, such as inflation and GDP growth, as opposed to changes in policy. The aggregate changes in housing subsidies due to changes in non-housing policies are very large. But not only have these implicit subsidies been very large, they have not been measured by traditional fiscal accounting measures. The present value of welfare losses is even larger; it may be as high as 6 percent of GDP.

Second, to discuss policies that would make these subsidies both less sensitive to changes in economic or financial conditions, and more transparent. Particular emphasis is placed on showing how the current system provides an implicit, and very inefficient, interest rate risk insurance program for homeowners. Explicit programs of this kind have been proposed in both the U.S. (see Kaufman, 1975) and Canada, (see Capozza and Gau 1984). The development of a well-designed insurance program could provide a significant improvement in the financial system's ability to allocate risks efficiently. Conversely, the maintenance of implicit schemes, such as the current approach, is likely to be both inefficient and inequitable.

1/ The features of the Argentine system which are common to other countries are: (1) the use of a tax fund to provide subsidized credit for housing; (2) the reliance on one of the largest depository institutions in the country to provide mortgage loans that are not completely indexed for inflation; and (3) real borrowing rates within the formal financial sector in excess of 10 percent p.a. In addition to Argentina such systems are in operation in Brazil, Colombia, Mexico, Peru and Turkey.

Finally, the paper provides a preliminary and seemingly paradoxical answer to the question: "Who should be first in the Housing Subsidy Queue?" In an economy as disrupted as the Argentine one, the answer is not the obvious one of those who have the most housing need, however that is defined. In this economy many high return housing investments are foregone because of policies that impede formal financial intermediation, increasing its cost and reducing its availability.[2/] Corresponding to this contraction in the availability of credit has been a similar contraction in the stock of housing. At the same time that the supply of housing has been contracting, real rents, like real borrowing costs, have also increased sharply. In many respects, Argentina's housing market appears to be functioning much like its financial system. However, unlike the effects of a shrinking formal financial system, i.e., increased consumption or capital flight, a contracting housing market can impose high costs on most lower-income families.

One way of reducing these costs is to target much smaller subsidies on those willing to mobilize their own resources. Such a subsidy distribution can be expected to have a much larger effect on net housing investment than does the current method of targeting very large subsidies on those least able to afford housing. The current subsidy approach redistributes wealth, but has little effect on output. Providing subsidies to those lower-income households willing and able to mobilize some resources could significantly increase the housing production induced by the subsidy. Perhaps double it or more. Such a large increase in production could lower rents for all renters rather than just the rents of the small portion of eligible families that the program can reach directly. Hence, it appears that the best way to provide the most housing assistance to the poor is not to target the subsidies exclusively on those who are the poorest.

A major reason why housing production could be expected to increase with a smaller per unit subsidy is because the subsidized housing units in Argentina are so expensive that few units are produced. According to a recent study for the Argentine Ministry of Health and Social Action <u>Informe Sobre El Sector Vivienda</u>, the average cost of housing units built by the main subsidy instrument was on the order of US$16,000 to US$18,000 in 1985. This figure compares with estimates of about US$2,500 to $3,000 per unit for Chile and Colombia for the same time period.[3/] Hence, it is clear that the average per unit cost in Argentina was a large multiple of the costs incurred in other countries during the

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2/ In Argentina between 1980 and 1985 broadly defined monetary holdings declined from over 22 percent of GDP to less than 13 percent. As is discussed further in the text, real borrowing costs also rose sharply. The figures on monetary holdings are from the <u>World Development Report</u>, 1987. Table 26.

3/ Source for Chile is Castaneda and Quiroz (1986); for Colombia it is in the Annual Report of Instituto de Credito Territorial, 1986.

same time period, and that the number of families served could increase sharply if lower cost units were produced.

Ultimately, questions such as: how many new housing units can be induced by a given government expenditure and, what is the effect of production on rent levels? are difficult empirical matters that are not answered in this paper. The purpose here is the more modest one of showing that it is important to consider, if not measure, the indirect effects that housing subsidies can have.

The plan of the paper is as follows. In the next section the Argentine economic crisis, the major housing finance institutions, and housing market conditions are briefly described. Then, in sections three and four user-cost estimates of the effects of various housing and financial policies are presented. A fifth section provides a measure of the resource costs to the economy of these policies and a measure of the scale of the implicit taxes that have financed the transfers. A six section describes the kinds of structural reforms of the housing finance system that could help reduce the costs to the economy of the current approach and a final section summarizes.

II. RECENT ECONOMIC AND HOUSING MARKET TRENDS IN ARGENTINA

A. The Economy

Argentina is a highly urbanized country with one of the most uniform distributions of income in Latin America. In 1980, 83 percent of the population was living in urban areas. Over the preceding 20 years approximately 11 percent of the population had moved from a rural to urban location. Its housing market is predominately owner-occupied, 62 percent, and one that in 1976 had just emerged from forty years of binding rent control. The late 1970's also saw the emergence of indexed mortgage loans. Prior to this time mortgage credit was provided at low fixed-rates, despite inflation rates of 20 to 30 percent per year.[4]

In recent years Argentina has also experienced a deep and sustained reduction in real income, very high real borrowing rates, the highest average inflation rate in the world for the 1975-1985 period, and

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4/ See Yujnovsky (1984), The National Housing Plan (1985) and Diaz-Alejandro (1970) for housing data, and a longer term perspective on the Argentine housing market.

capital flight of significant scale.[5/] Over the 1975-1984 period, real per capita income fell by more than 20 percent to approximately $2,100 per capita. The size of the formal financial system contracted sharply with the extension of regulated interest rate ceilings, real borrowing costs have been higher than 30 percent per year, deposit rates were deeply negative over much of this period, and net investment and particularly housing investment, was very low, the latter accounting for less than 3 percent of GDP.[6/]

B. The Housing Sector

Against such a background it is not surprising that the public sector share of units of housing produced almost doubled (going from 28 percent in 1979 to about 50 percent in 1985), as the unsubsidized demand for housing should certainly fall in such an environment. Nor is it surprising that long-term credit should disappear from such a risky environment, and long-lived investments such as housing should be deferred. What is surprising is the level of disruption in the housing market. This market is in complete disarray. The flow of net new housing production has been very low. For the 1982-85 period it was significantly below the rate of household formation expected by the growth of population and may well have been negative.[7/] In addition to a limited flow of new production, the services provided by the stock of existing housing contracted. For example, in 1985 the supply of housing units for rent in Greater Buenos Aires, a market that accounts for over 40 percent of the national rental market, was 25 percent less than the 1980 figure. Owners apparently felt that in the uncertain economic environment that unoccupied units were preferable to ones with sitting tenants. During this same time

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5/ Khan and Ul Hague (1987) show that over the 1974-82 period that of all the major debtor developing countries, Argentina had one of the highest levels of capital flight. The data on Argentina are from various published World Bank Reports, The World Development Report and the Central Bank of Argentina. Computations on inflation from International Financial Statistics, December 1986, International Monetary Fund, Washington, D.C. The real borrowing rate figure is from Boschen and Newman (1986).

6/ Source: For housing investment data, the Argentine Housing Secretariat. See Comparative Housing Production in the Industrialized Countries, OECD, Paris, 1982 for data on the levels of housing output in various countries. The Argentine level over 1982-85 was 3.0 housing units produced per thousand population, a very low level by comparison with countries of similar income levels. As discussed in the Appendix, housing production was also low on an historical trend.

7/ On the rate of household formation see the UNDP study cited in footnote 3 and the Appendix. It exceeded production by more than 40,000 units in 1985, after having exceeded production levels by an average of 15,000 units per year over the previous three years.

period, real rents for housing units in Greater Buenos Aires more than doubled.[8/]

C. The Housing Finance System

Almost all housing finance in Argentina is provided through two intruments: an earmarked wage tax fund, the National Housing Fund, FONAVI, and a National Mortgage Bank, BHN. Since the late 1970s public housing has been funded mainly by FONAVI. In 1986, the tax was estimated to yield the equivalent of about 700 million australs, about 1.4 percent of GDP. The tax has provided funding for over 70 percent of subsidized housing produced over the 1980-85 period. FONAVI beneficiaries had an average per capita income of about US$750, significantly below the per capita income level of US$2,100, and the housing units produced cost 18,300 australs or almost U.S. $17,000.[9/]

Mortgage payments for FONAVI beneficiaries are made for a term of such length that payments are a small share of income and a zero real interest rate is used to amortize the loan. Payments are indexed to minimum wages rather than the inflation rate or cost of funds, and loans are for the full amount of the cost of the house. Households do not make any downpayments.

FONAVI's ability to recover resources is hampered by poor recovery of the value of the money lent to provincial housing institutes (IPVs). These institutes identify contractors and subsidy beneficiaries, pay contractors in accordance with the costs of production, collect mortgage repayments from borrowers and repay FONAVI. The IPVs have an 18 month interest-free grace period before their payments to FONAVI are begun, and their penalties for payments that are later than 18 months are trivial. Consequently, even though households tend to make repayments, because of the high rate of inflation only a small fraction of the real

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8/ Longer term figures on housing production are from Yujnovsky (1984) and the National Plan which relies heavily on census statistics. The figures on rents and the number of units offered for rent are from Informe Mensual de Alquileres, Direccion Nacional de Investigacion y Desarrollo, Julio, 1986. See The Appendix for a further discussion of housing market trends.

9/ The estimate of FONAVI tax collections is from the UNPD Study cited in the text, Informe Sobre El Sector Vivienda (1987). The figures on FONAVI housing production and the income of beneficiaries are from discussions with Housing Ministry (SVOA) staff. The FONAVI housing cost estimate is from the UNDP study and corroborated by The Inter American Development Bank in 1985. Australs were equal to 1.1 to the US$ in late 1986.

value of FONAVI expenditures, on the order of 2-5 percent, is ever recovered.[10]

The National Mortgage Bank (BHN) is the chief alternative source of funds. In recent years BHN increased its output. At the beginning of 1987 it was third largest financial intermediary in the country with a US$1.8 billion portfolio, almost completely comprised of indexed mortgage loans. By that time, BHN had evolved from being a bank that had generated about half of its funds from internal sources during the 1970s to one that had a negative cash-flow that was maintained by a US$1 billion in loans from the Central Bank of Argentina and the deposits of public agencies.[11]

BHN's loan terms are typically for 80 percent of house value, and carry an initial average contract real interest rate of about 3.5 percent. Lower-income families pay 1 percent per annum, increasing to 5 percent for those with higher salaries. The deterioration in its financial position occurred for two reasons: (1) partial indexation on its loan portfolio has significantly reduced the real value of older loan repayments;[12] and (2) it increased the volume of its lending despite an inability to repay its outstanding loans to the Central Bank.

There is almost no other credit available for house purchase. For example, almost all housing sales in Buenos Aires are made with cash in U.S. dollars. In those instances in which households cannot generate sufficient dollars from their own savings, they can borrow through schemes such as those operated by the Provincial Bank of Buenos Aires at

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10/ Source: UNDP Study corroborated by SVOA.

11/ Source on data for BHN is from the UNDP Study and corroborated by the Central Bank of Argentina. The public deposits have accounted for 85 to 90 percent of the Bank's deposit base, and the assistance from the Central Bank accounts for 20 percent of the Central Bank's quasi-fiscal deficit.

12/ A number of different indicies are used to index repayment of BHN loans, but none of them completely insulate the value of the outstanding loan from changes in the rate of inflation or interest rates. The most common index is one that provides for payment increases based equally upon the behavior of salary and price indices. In a situation in which real wages fall, this kind of index will cause the value of the loan to be discounted, decreasing the effective yield. Of course in such circumstances, the decrease in value is smaller than the decrease in value of FONAVI loans which are indexed only to wages.

inflation-adjusted rates that in late 1986 were in excess of 30 percent.[13/]

To summarize, housing market data suggest a market in disequilibrium: sharp and sustained increases in real rents and real rent to value ratios do not elicit positive net new production, and for much of the period they are associated with a contraction in the supply of existing housing. Significant resource transfers to the sector through FONAVI and BHN (1.5 to 2 percent of GDP) appear to induce little incremental private expenditures on housing. The next section discusses how housing and financial policies interact to sustain this disequilibrium, and how the sustainment of this disequilibrium affects the efficiency of the resource transfers to the sector.

III. A USER COST OF CAPITAL APPROACH TO THE HOUSING MARKET

To evaluate the costs of a disequilibrium of the housing market and the way the form of the housing subsidy affects it, what is meant by disequilibrium must first be defined. I assume a disequilirium occurs if the present value of a housing unit, Pk, is greater than the cost of the house and land, Pc, for an extended period of time. The FONAVI program can be defined in these terms as a "large" in-kind capital grant to lower-income households.[14/] As a capital grant, it represents a subsidy almost equal to Pk, the value flow services provided by the housing unit. In a market not experiencing disequilibrium values of Pk in excess of Pc would induce additions to the housing stock and vice versa. As mentioned above, it does not appear that the Argentine housing market is one in which suppliers have responded to price incentives. However, because the price of FONAVI units are a function of costs, Pc, rather than the valuation of the resources, we cannot directly observe Pk or this discrepancy.

In order to determine how policies have affected Pk we consider how the determinants of the unobservable Pk have charged over time. For example, if Pk were determinely solely by taking the present value of expected rents, as is assumed by Kelley and Williamson (1984), then assumptions about the expected real rents and the equally unobservable real discount rate would yield a measure of Pk. The fact that the real interest rate is not observable complicates any such computation. But an even more important computational problem arises because housing provides more than just shelter services. Like other forms of capital it also

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13/ Source: Provincial Bank of Buenos Aires.

14/ It can be viewed as a "large" capital grant because of the value of the house relative to family income. Inclusive of land the FONAVI units were worth about $22,000 in 1985 whereas family income was on the order of $4,000. The ratio of these figures is more than double the ratio that exists in most countries for privately initiated house purchases.

interest rate is not observable complicates any such computation. But an even more important computational problem arises because housing provides more than just shelter services. Like other forms of capital it also provides a store of value for savings, a service that plays an increasingly important role in evaluating the behavior of housing investment under inflationary conditions. Unfortunately, not only is this type of return also not observable, it depends upon the returns to the alternative savings instruments available.[15]

With all the unobservable components of costs and returns we are left with a computation for the present value of FONAVI housing units that must be described as a stylized figure. Neverthless, for the purposes of this study, computations of these stylized measures of present value, or their flow analogues, user-cost measures, have a number of significant advantages. First, the user cost approach, which calculates the cost of having the full use of the services of the asset for a particular period of time, underlies almost all empirical studies of investment behavior.[16] Consequently, the construction of such stylized figures for the housing sector is simply an extension of empirical investment analysis to housing investments. Indeed, almost all recent empirical analyses of housing investment use such an approach.

15/ Under this broader definition of the services provided by housing, the present value of housing services is equal to

$$V_o = \int_o^T [a(t)] \, e^{-Rm} \, dt + [V_T] \, e^{-RT}$$ Where V_o is the value of the

investment in time 0, and a(t) is the imputed value of the stream of housing services yielded by the property in year t. This latter figure is net of repair expenses; R_T is the real discount rate, R_m is the mortgage rate. The first bracket measures the net flow of rental services. The second bracket the value of the investment at a time T. See von Furstenberg (1977) for a fuller discussion of this approach and the way property taxes, depreciation of structures, inflation, and tax considerations can enter the analysis.

16/ Branson (1976) provides one of the most lucid short discussions of the user cost concept and its relationship to present value. As he says the inability to measure the cost of capital services "is the major difference between the market for capital goods and that for other inputs, ... that makes capital theory so complex and makes measurement of capital inputs very difficult," p. 214. The user cost or rental value of services expression can be derived from the present value equation given in footnote 15. It measures costs on flow basis rather than a stock basis as does a present value calculation. It shows that the cost to be paid for access to an asset is equal to the net rent less the expected appreciation.

the financial system, the development of such measures are prerequisite to the control of government's command over resources and risk exposure. In this respect, the approach is very similar to the new budgetary measures applied to the U.S. credit budget, and the demonstration of the applicability of the technique is more important than the particular estimates made.[17/]

Third, empirical studies of Argentina by Diaz-Alejandro (1970), and Mallon and Sourrouville (1975) both argue that one of the most serious failings of government policy in Argentina has been the way that various policies have increased the relative price of the existing capital stock. They estimate that if capital formation figures through the 1960s were adjusted to reflect the relative prices of capital goods that these figures would be reduced by as much as 30 percent. Based on their empirical work they conjecture that these relative price increases are due to licensing requirements, import tariff restrictions, and mortgage credit subsidies. Applying a user-cost of capital perspective to the housing sector is very much in the spirit of these analyses. Indeed, its application is the kind of extension of this earlier work that is necessary to give their empirical conjectures analytic and policy content.

Dougherty and Van Order (1982) have used the user cost approach to show the identity between the return a homeowner would receive and the rent that would be charged by a landlord under competitive conditions. de Leeuw and Struyk (1975), among others, have emphasized the potentially important effects that various qualities of housing investments could have the linkages between the markets for the stock of housing and the flow of housing services. They argue that the lumpiness of housing expenditures usually necessitates borrowing for purchase. However, lenders' decisions to provide financing usually have little to do with the expected future returns to the asset financed housing. It is the prospects for household income rather than the return to the investment that determines whether a loan is made. Consequently, even though a housing investment may be expected to have a high real return, if the prospects for wages are not as optimistic no loan is made. With such underwriting requirements it is easy for difficulties to arise in undertaking a high yield investments when income growth prospects are uncertain. This difficulty in borrowing,

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17/ See the Special Analysis of the Budget, Office of the President, Office of Management and Budget (1986) for a discussion of the recent attempts to measure the cost to government of various credit programs. It is based on a user-cost concept. See Break (1982) for a discussion of issues in measuring the cost of government credit programs.

in turn, can produce a lasting discrepancy between housing's cost and the capitalized flow of the value of housing services. In fact, this type of disequilibrium-rationing perspective has been the basis of virtually all econometric analyses of the short-run behavior of housing and credit markets in both developed and developing economics.[18/]

IV. APPLYING THE USER COST APPROACH TO THE ARGENTINE HOUSING MARKET

I suggest that the Argentine housing market has been characterized by such a disequilibrium. I then follow the approach used by Laidler (1969) to explain how various policies have affected housing's asset price.

For example, for a house costing 18,000 australs to produce, situated on land worth approximately 6000 australs, as is the case for FONAVI units, assume that the mortgage payments of beneficiaries are equal to 20 percent of household initial income of 4,000 australs per year, or 800 australs per year. Assume also that payments are maintained in real terms by loan indexation and paid in a timely fashion. In this case, the present value of the repayments of the FONAVI loan would be equal to 10,500 australs. The difference between the present value of the loan repayment and the value of the house is the amount of subsidy the household receives. It is equal to about 13,500 australs, 57 percent of the value of the house because FONAVI pays for the structure, but not the land, which is contributed at zero cost. It gives a net subsidy equal to 43 percent (18,000-10,500) of its expenditures, and 54 percent (7,500/13,500) of the net transfer the household receives.

These levels of subsidy would apply if the present value of housing services, Pk, was approximately equal to the of production and land, Pc, i.e., cost around 24,000 australs. Such a result could occur, for example, if (1) net rents were equal to 8 percent of house value per year for 30 years, (2) housing's real net value were expected to be constant, and if the real interest rate were 4 percent, and administrative costs of managing and collecting mortgage payments were 2.5 percent per year. In this case the rental payment stream would be discounted by

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18/ See Buckley and Gross (1985) for a review of this literature in developed countries, and an analysis of how financial reform can affect this kind of disequilibrium. See Kelley and Williamson (1984) for the application of this type of model to housing investments in developing countries.

6.5 percent, and the store of value at 4 percent.[19/] The house under these assumptions be worth 24,400 australs; 18,850 australs of this value would derive from the flow of rental services, and 5,550 australs the present value of the store of wealth.

These assumed equilibrium housing market conditions are obviously very different from the Argentine housing market conditions described earlier. Hence, the disequilibrium affects the size of the transfers. Now consider how various aspects of Argentine economic policy affect the disequilibrium.

A. The Effects of High Real Interest Rates, and Increasing Real Rents on Housing Valuation

Although real rents in Greater Buenos Aires have been increasing sharply and almost continuously for a long period of time, the net supply of housing in this market has contracted. In addition, as discussed earlier, and more fully in the appendix, in recent years net new housing production in the entire country has been very low and probably below zero. On the one hand, such low production levels are not surprising when one notes that real borrowing costs in the Argentine financial system exceed 30 percent. Indeed, if a 30 percent discount rate were used to evaluate investment decisions, even the 16 percent per annum real increases in rents over the 1980-85 period would result in disinvestment in housing. Consequently, the observed negative growth rate of housing capital would not be surprising.

However, as documented by Khan and Ul Hague (1987), significant amounts of Argentine capital have fled the country to invest at real international interest rate that, according to Hendershott (1985), ranged from 2 to 5 1/2 percent during this period. Besides making investments abroad, Argentines also made dollar-denominated investments in non-subsidized housing within Argentina. The opportunity cost of these investments is the real international interest rate less transaction costs. As a result, if there were credible formal financial intermediaries

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19/ The rationales for assumptions made about the unobservable components of the present value calculations i.e., the real interest rate, and the administrative costs of mortgage intermediation are described in the text below. The bases for the observable components i.e., net rent to value, land values, and real house prices are based on discussions in The Housing Ministry, various provincial housing authorities and the newspaper survey of rents and prices carried out by the Housing Ministry. The imputed rent to value ratio is consistent with the empirical work of Malpezzi and Mayo (1987), and similar to the assumptions made by Laidler (1969). It is lower, and therefore more conservative, than the ratio that can be inferred from the Kelley and Williamson (1984) analysis. The assumptions also, of course, produce estimated values of Pk that are approximately equal to Pc. Hence, they represent what might be termed market equilibrium assumptions.

in Argentina, funds could be borrowed at the same real interest rate at which Argentines are investing them abroad (plus fees for loan repayment and loan servicing which are fairly stable).

In this perspective almost all of the more than tripling of real borrowing costs observed in the formal Argentine financial system reported by Boschen and Newman (1986) is the result of macro-financial policies which increase the costs and risks of intermediation in the formal financial system. The higher costs are, in effect, selective credit policies or taxes that drive tax wedges into the evaluation of the value of investments in different sectors of the economy. The size of these wedges is in direct proportion to sector's reliance on the formal financial system. Whether such policy-induced increases in real interest rates stem from exchange rate policies or policies to support insolvent domestic financial institutions or from other policies is an important, and perhaps intractable question. See, for example, Edwards and Khan (1985). Fortunately, it is also one that is beyond the scope of this paper.

Here, the focus is on examining the effects of the high real borrowing rates on housing investment incentives. This focus is maintained by first making a conservative estimate of what the "pre-tax" resource value of a housing investment would be if financing were available that reflected the cost of funds rather than the costs of intermediating in Argentina; second, comparing this figure with the observed cost of such an investment. The difference between these figures provides a present value measure of the profitability of housing investment that is foregone due to policies which prevent or obstruct savings from moving to high return investments.

To simplify, I focus on portfolio investment decisions rather than investment versus consumption decisions, and ignore reinvestment risks. This focus permits housing investments to be compared with a composite alternative investment of similar risk. I also assume that the expected returns to housing are determined by a distributed lag process that takes into account past increases in rents and real interest rates and causes households to expect to have annual rents increase from 8 to 10 percent of house price.[20/] Finally, I assume that any increase in the

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20/ Specifically, I assume that the 16 percent annual real rent increases observed in Greater Buenos Aires over 1980-85, the declining housing opportunities described in the appendix, and the roughly 3 1/2 percent increase in real international borrowing costs noted earlier, cause households to expect to have to pay annual rents equal to 10 percent of the cost of the unit. According to discussions with housing officials this assumption is conservative. It is also of course a very simple assumption. Again, see Kelley and Williamson (1984) for a discussion of how expectations can affect such investment decisions. They assume naive expectations.

administrative costs of intermediation that may have occurred (as, for example, when staff cuts at intermediaries are not made in accordance with a contraction in the volume of funds mobilized) was the result of selective credit employment policies. Hence, any change that may have occurred in this cost of intermediating does not affect the pre-tax resource value of housing investments.

With these assumptions, households who gain access to a FONAVI house receive housing services, a wealth transfer, and a forward financial transaction. The last permits households to avoid expected future real rent increases. A present value of expected rent increases of about 4,700 australs can be added to the 18,850 australs value of rental services flowing from an 8 percent annual net rental value. The value of avoiding the expected rent increases, i.e., the forward transaction, is equal to 19 percent of the value of the housing unit.

The estimated total resource transfer in this case is 29,100 australs. Subtracting the present value of the mortgage repayments indicates that the household receives a net resource transfer equal to around 18,600 australs. This amount is more than 100 percent of the assumed FONAVI expenditures, even though by fiscal accounting measures FONAVI provides a subsidy that is equal to only 42 percent of its expenditures. These effects are shown in bar II in Figures 1 and 2.

Higher borrowing costs are not the only implicit tax imposed on households by credit policy. Besides having to pay more to borrow, they also receive negative returns on their savings as long as the saving is held in financial assets. This kind of regulatory tax also affects the value they place on housing as a store of wealth.

B. The Effects of a Repressed Financial System on Housing Valuation

Between 1978-1985, real housing values in Greater Buenos Aires oscillated gently around a rising trend.[21/] Few other forms of domestic saving maintained their purchasing power during this period. Hence, housing has been a relatively attractive form of domestic savings. Suppose that interest rate ceilings on deposits interacted with inflation to reduce households' discount rate in such a way that the after-tax-real

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21/ Source: SVOA.

FIGURE 1 : PRESENT VALUE OF A FONAVI HOUSE AND MORTGAGE

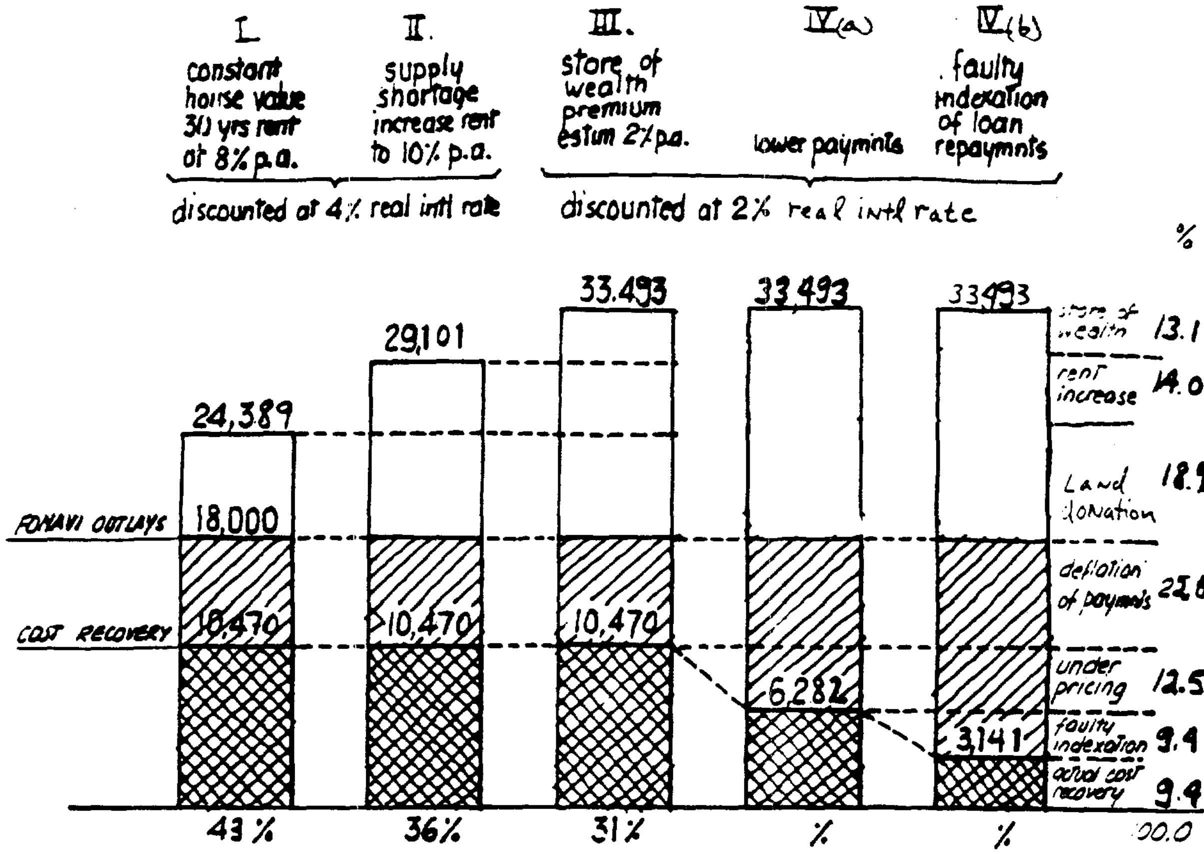
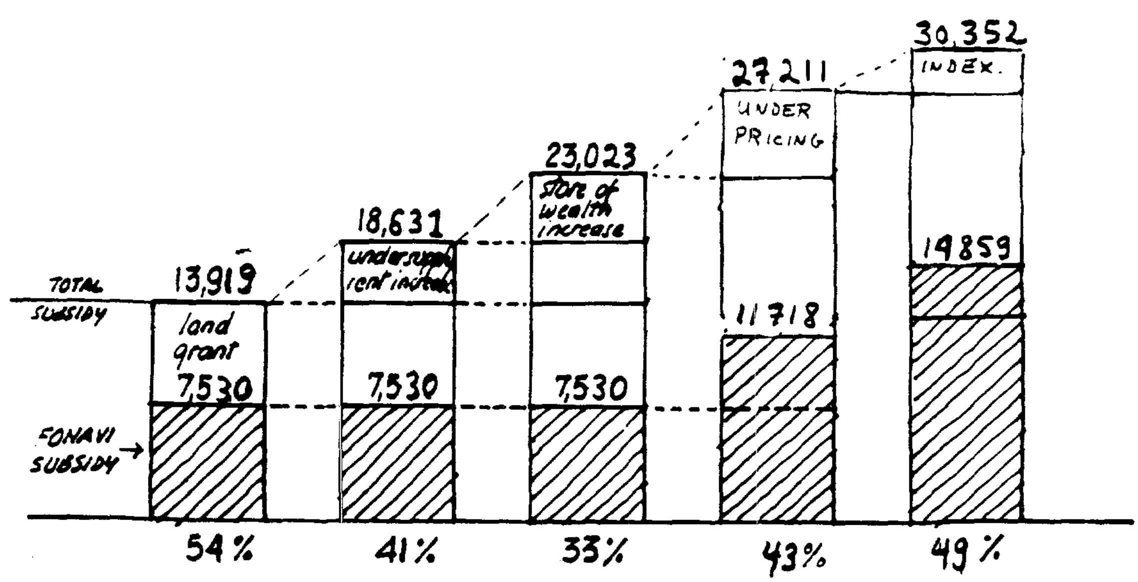


FIGURE 2 : FONAVI SUBSIDY AS A SHARE OF TOTAL RESOURCE TRANSFER



Measured in austials which in 1986 were equal to 1.1 to the US$.

opportunity cost of funds was 2 percent rather than 4 percent as was assumed earlier.[22/] Under this change in assumption, the present value of the house increases to 33,500 australs. The store of wealth value increases from 5,550 australs to 9,900, and all other components of value and repayment remain the same. The net resource transfers to households the FONAVI subsidy (18,000-10,500) measures about one-third of the subsidy the household receives (33,500-10,500). The per unit subsidy increases by almost 9,000 australs relative to what it would be in housing and credit markets that were not in disequilibrium i.e., Case I. In effect, the interaction of inflation and interest rate ceilings has increased the steady-state demand for housing as an asset of refuge. Simultaneously, through increasing borrowing costs to extraordinary levels, credit policy has reduced the ability of the economy to fulfill this demand.

C. The Effects of the Subsidy Program Structure.

Ex Ante Subsidy Problems

To this point it has been assumed that FONAVI loan repayments are indexed for inflation and are equal to 20 percent of initial household income. In fact, initial payments have averaged only about 12 percent of income. As a result, the ex ante per unit subsidy is larger. Due to this change the present value of mortgage repayments to FONAVI falls from 10,500 australs to 6,300 australs, and the FONAVI subsidy rate goes from about 40 percent of its expenditures to more than 65 percent.

This larger per unit subsidy allows lower-income households to afford payments. In principle, at least, it enables the subsidy to be targeted towards those with the most housing need. As a result, of course, fewer of the many eligible households can be helped. But, before addressing this important targeting problem another important subsidy must first be identified.

Ex Post Subsidy Problems

Although not emphasized in Figures 1 and 2, one of the more important subsidies is given through the repayment index scheme. FONAVI mortgage payments are indexed to wages. As a result, the lender, i.e., FONAVI, bears real income risk. When real income falls, mortgage payments do not increase as rapidly as inflation. In many cases payments now often

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22/ This is similar to the approach taken by Barro (1979) in his discussion of effects that capital market imperfections might have on the evaluation of government debt. However, in this case the focus is on the portfolio effects with respect to one asset rather than total wealth. The assumption appears to be conservative given that Diaz-Alejandro (1985) shows over the 1980-1984 period the average real deposit rate was negative and there are significant transactions costs to capital flight, particularly for small amounts of capital.

represent a tiny share of income. For example, the average loan repayment received by FONAVI in 1985 was equal to 1/5 of 1 percent of house value, or 36 australs per year. In effect, the repayment indexes "over-protect" already heavily-subsidized households from the interest rate risks of the macroeconomic environment.

Providing some form of insurance or protection of this sort for borrowers may be a reasonable strategy in economy where real income growth is uncertain, and interest rates have reached such high and volatile levels. However, the Argentine approach is very expensive. The indexes and repayment terms used by FONAVI provide more than complete insulation from interest rate or real income risks. Under its scheme the small number of households who are lucky enough to gain access to housing assistance are not only insulated from economic shocks, their position has improved substantially when economic growth has faltered or inflation increased.

The repayment scheme effectively provides real-wage insurance for income reductions for the more than 200,000 families with outstanding FONAVI loans.[23/] Unfortunately, the benefits of this insurance become operative regardless of the behavior of household income. If, for instance, a FONAVI beneficiary experienced an increase in real income at the same time that there was a reduction in the aggregate minimum wage index, his real payments would still be reduced because of the index used. This indirect form of insurance against reductions in real wages is not targeted on those who need it.

To approximate the effects of partial indexation on the size of the per unit subsidy assume that throughout the life of the loan that real payments are expected to decline in such a way that the average payment is half of what it would have been if indexation adjusted for changes in the inflation rate. As a consequence, expected annual payments are equal to 6 percent of initial income rather than 12 percent, or 240/18,000 = .013 of house price per year. This figure, 1.3 percent of house price, is 1.2 percent lower than the 2.5 percent per year administrative costs of operating the program.

In this case, the subsidy is equal to: (1) the entire imputed rental value of the housing units, since the mortgage payments do not amortize a zero interest rate loan; (2) the store of wealth remaining after the house has been paid off; which is relatively high because there are few other savings options; and (3) a portion of the administrative costs of operating the program. Such a program structure must finance current costs out of either future repayments or it must continually contract in size.

Bars IV (a) and (b) show the incremental resource transfers and subsidy levels implied by the higher per unit subsidy level and the assumed indexation scheme. The net resource transfer (33,500-3100) is 169 percent

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23/ Source: On the total number of FONAVI loans, the UNDP study (1987).

of FONAVI expenditures (18,000 australs), and the measured FONAVI net subsidy (18,000-3100) equals about half of the net transfer received.

The misleading nature of subsidy evaluation using what might be termed a traditional fiscal accounting concept is particularly striking. Such a traditional measure would suggest that the FONAVI subsidy is equal to 83 percent of its expenditures (18,000-3100)/18,000. This is a high subsidy rate, but it is also one that is self-contained, in the sense that revenue collections exceed net expenditures. The "implosive" nature of the subsidy structure is not apparent. By decomposing the subsidy into its various components, the user-cost perspective helps to show that the program is not self-contained; the repayments do not even cover the costs of administering the program.

V. EVALUATING THE EFFICIENCY OF HOUSING SUBSIDIES

The finding that the value of the transfers given to FONAVI beneficiaries (33,500-3,100) have been almost double the amount that is given through FONAVI's expenditures (18,000) suggests that these benefits have been financed through two sources: the explicit tax revenues collected by FONAVI, and implicit taxes on non-beneficiaries through inflation and relative price effects. It also suggests that the implicit taxes have been about the same scale as the amount raised through direct taxes, in aggregate over 600 million australs per year.[24/]

A. Expenditure Efficiency

Housing finance subsidies are distributed in a housing market that does not clear, and the resource value of a new unit is worth at least 30 percent more than the cost of supplying the good. While this disequilibrium is the result of the interactions of macropolicy and credit regulations rather than housing policy, the latter policy is also flawed because it does not attempt to reduce the economic losses due to the

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24/ In addition to FONAVI's implicit subsidies, BHN loan indexes have also caused outstanding loan values to be discounted as a result of changes in the inflation rate or real wages. Hence, these loans also contain an ex post off-budget subsidy in addition to an implicit ex ante subsidy. BHN's ex ante per unit subsidy is about 45 percent if the same 6.5 discount rate as the FONAVI loans is to discount repayments and BHN's initial 3.5 percent real coupon rate as the cost paid by the borrower.

disequilibrium. Instead, it distributes very large subsidies that are based on poorly-targeted measures of need.25/

Most of the subsidy is distributed to what are termed inframarginal households, i.e., those households whose behavior is unaffected by the changes in market conditions. It is given to those who already have outstanding loans, (both FONAVI and BHN), or those so poor that new housing was unaffordable before the change in macroeconomic conditions (through FONAVI). As a result, the subsidy distribution avoids the more traditional problem of efficiency losses caused by subsidies changing prices, see Break (1982). But, by ignoring the marginal households that have been rationed out of the housing market, the subsidy does not help cushion the sector or the economy from the effects of these policies. For example, suppose that in the long-run housing can be produced under constant return to scale, but that in the short-run production responds to the price of the existing stock. Whenever the price of the existing housing stock exceeds costs investment in the sector takes place. (See Kearl (1978) for the development of such a model.) However, if production does not respond to these price incentives disequilibrium of the sort I have hypothesized occurs.

In Figure 3 the observed stock of housing, q, represents the intersection of the constrained demand curve for the stock of housing, curve d'd' with the existing stock of housing. This curve is below the demand curve, schedule dd, that could be achieved in a market not in disequilibrium. Suppose further that the FONAVI and BHN subsidies are distributed inframarginally. They shift d'd' to d"d".26/ At the

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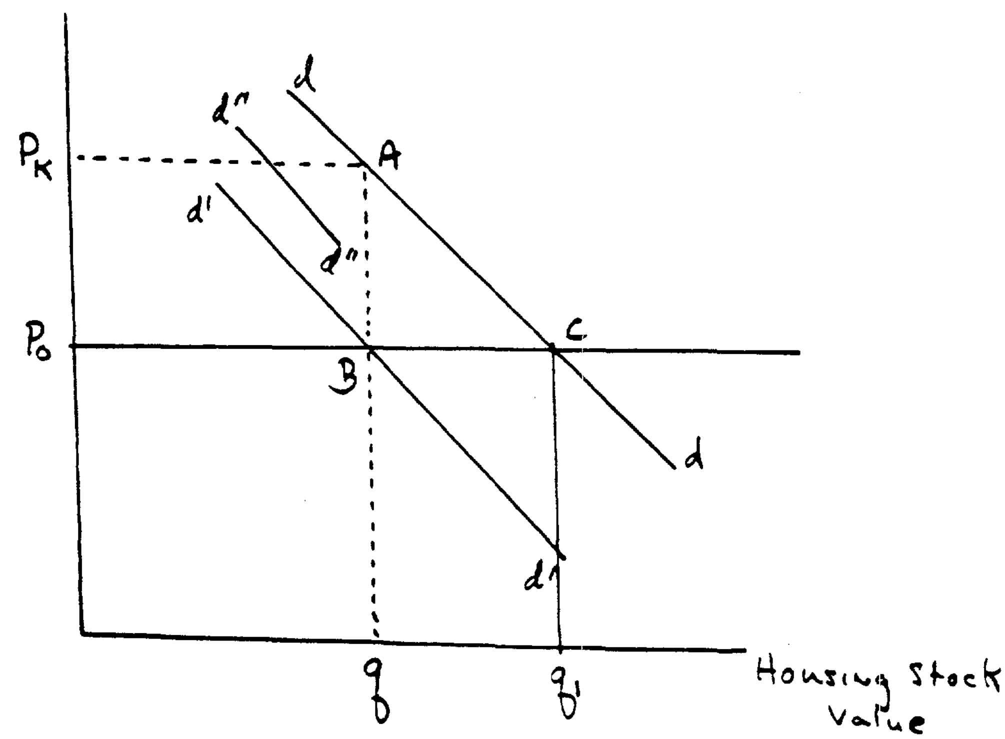
25/ For example, although Buenos Aires is the most rapidly growing area of the country and contains 40 percent of the housing stock, it receives only 20 percent of FONAVI funds. In determining how funds should be allocated among cities FONAVI uses housing stock characteristics which may be associated with a declining market to direct resources. As a result, it may direct resources away from markets with the greatest demand.

26/ See von Furstenberg (1976) for a discussion and estimation of the share of a U.S. housing finance subsidy that goes to inframarginal households. His results indicate that over 85 percent of the subsidies were inframarginal and did not induce new production. They are consistent with Murray's (1985) empirical fundings for the U.S. Because of the structure of the subsidy a smaller share of the Argentine subsidy recipients are likely to be marginal new house buyers than was the case in the U.S. program. Hence, the assumption of zero distribution of subsidies to marginal buyers, while obviously the lower bound of how many could have been affected, does not seem too unreasonable.

constrained output level, the short-run price, Pk, exceeds Po, the long-run equilibrium price. There is no deadweight loss implied by the subsidy's distribution; marginal incentives have not been affected by it. However, the efficiency loss associated with the disequilibrium, area ABC, has also not been affected by the subsidy's expenditure.

FIGURE 3



A linear approximation of area ABC is 1/2(Pk-po)x(q-q1). This figure can be approximated by an estimate of the difference in the present value of a housing investment, Pk, relative to the costs of production, i.e., 37 percent, an estimate of the elasticity of housing demand with respect to price, and an estimate of the value of q, the value of the Argentine housing stock. If the housing stock is equal to 100 billion australs, as can be inferred from calculations in <u>Plan Nacional</u>, and the absolute value of the price elasticity is on the order of .75, then the present value of the deadweight loss is 4.9 billion australs or about 6 percent of GDP.[27]

Besides the deadweight loss that is not addressed by the subsidy's distribution another major inefficiency in the subsidy occurs because of the in-kind nature of the grant. Aaron and von Furstenberg (1972) have shown that, under representative assumptions about household preferences, a per unit subsidy rate of 80 percent, would lead to an efficiency loss of 50 percent of the subsidy. In other words, if households were given a cash grant equal to half the amount given to them through the in-kind transfer they would be equally well off. Because the FONAVI subsidy rate, even without the disequilibrium increase in the value of Pk, is in excess of 80 percent, one might also expect very large efficiency losses due to the subsidy's in-kind structure.

If the subsidies could be directed to those rationed out of the market they could shift the d'd' schedule towards the dd curve. Such a shift would reduce the efficiency losses from the sector's disequilibrium.[28] Reducing the per unit subsidy size from the current FONAVI subsidy levels of more than 15,000 australs per unit to a much smaller lump-sum subsidy could be expected to induce significant resource mobilization by households. Similarly, reducing the ex post subsidies given by BHN to those who already own homes could help discriminate between borrowers who are willing to pay for a house and those borrowers who are counting on loan forgiveness to pay for it.

27/ See Malpezzi and Mayo (1987) for housing price and income elasticity estimates for developing countries. The calculations here assume that the slightly fewer than 9 million Argentine housing units are worth 11,000 australs each. The observed q of 100 billion australs is 27 billion australs less than the steady-state level of q1. The figures on the share of wealth in housing and its relationship to GDP are consistent with similar figures for other countries presented by Goldsmith (1985), and similar to the stylized figures of representative developing countries estimated by Kelley and Williamson (1984). Therefore, 1/2 (.37) (27b) = 4.9 billion australs. A 20 percent reduction in the value of the housing stock from its steady state non-taxed level may appear to be very high since housing output in any given year can change the stock by only 2-3 percent. However, the Argentine housing market has been in disequilibrium, for an extended period of time, and a lack of availability of mortgage finance has probably been capitalized into lower prices for the entire stock.

28/ The deadweight loss is presented in present value rather than annual terms because unless there is a policy change the annual loss will continue to be realized. See Boskin (1978) for a discussion of such present value estimates.

Such a change in approach would allow household savings rather than the government's per unit transfers to increase as macroeconomic conditions increased the resource value of housing. This is exactly the approach that has been implemented in Chile. Smaller up-front housing subsidies there, of on the order of U.S. $2,500 per unit, and not more than 75 percent of unit cost, leverage household savings. (See Castaneda and Quiroz, (1986) for a discussion of the Chilean housing subsidy scheme.) The result is that government expenditures should be able to induce more units of housing production per dollar of expenditure. A large increase in production should lower rents. Over time the increase in production could also help to reduce the implicit taxes on household saving. The following table shows the effects that reductions in these implicit taxes could have on the deadweight losses that occur in the housing market.

The Present Value of Deadweight Losses Implied
by Combinations of Housing Rent/Value
Ratios and Household Discount Rates

(In billions of 1986 australs)

	Discount Rate		
Rent/Value	.01	.02	.03
.11	8.1	6.2	4.9
.105	7.5	5.7	4.3
.10	6.8	4.9	3.7
.095	6.1	4.3	3.0
.09	5.7	3.8	2.3

The situation described by Figures 1 and 2, bar IVb is depicted by the middle of the center column. The higher rents and lower after-tax discount rates result in a long-run deadweight loss of 4.9 billion australs. The effects of increases in production which lower rent-to-value ratios can be read by going down columns; the effects of reducing the taxes on savings can be read across rows. These figures are clearly meant to be only illustrative. Nevertheless, they show that in Argentina focusing such large per unit subsidies on the poor who are presumed to be unable to mobilize any resources of their own, rather than on those among the poor who can and will mobilize resources is very expensive. They also show that under less conservative assumptions about the effects of credit policies on rents and household discount rates that the present value of the deadweight losses could be much larger.

B. Revenue Efficiency

Now consider the inefficiencies of the revenue mobilizing process. These implicit housing subsidies have been financed with similarly implicit taxes on other sectors of the economy. These implicit taxes have often been imposed on resource owners who are better able to avoid taxes than are those whose income is within the formal tax base and explicitly taxed. By funding housing subsidies through, for example, inflation and the negative returns to savers in financial instruments, Argentine fiscal policy has almost certainly tried to tax some of the most price sensitive inputs in the economy. By attempting to impose large implicit taxes on those most able to avoid them, tax policy has generated large efficiency losses.

Some notion of how large these implicit taxes are can be inferred from noting that since 1980 FONAVI and BHN have issued mortgages that at the time of loan origination were worth about 4.2 billion australs in 1985 prices. The present value of these loans is now on the order of 1 billion australs. Hence, mortgage debt forgiveness of about 3.2 billion australs, an amount equal to 25 percent of all the financial assets in the economy in 1985, was implicitly transferred to homeowners.[29/] While it is difficult to identify the incidence or effects of such a tax, it is even more difficult to argue that it is not of significant scale.

VI. RESTRUCTURING HOUSING POLICIES

The list of problems with the Argentine housing finance system is a long one. First, both BHN and FONAVI's subsidies are distributed in ways that do not address the disequilibrium in the housing market. As a result, housing production has fallen and, real rents have risen more than they would have if the subsidies were distributed differently. While economic circumstances may have dictated a need for a reduction in investment in the sector, they did not dictate that the subsidies ignore the welfare losses that they could have helped reduce. Second, housing costs for the large number of poor who cannot possibly be served by the FONAVI program appear to have increased rapidly. Not only is this trend regressive, it also contributes to problems with stabilization of the economy. Rent increases of the scale that have occurred are almost certainly an important contributing factor to an inflationary policy that has produced contemporaneous real minimum wage increases despite a steep decline in per

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29/ Source: On volume of financial assets, IFC statistics. The estimate of loan forgiveness is derived by capitalizing the current mortgage repayments to BHN and FONAVI at a 6.5 percent rate. The original loan amount equals the average loan times the volume of lending since 1980.

capita income.[30/] Third, the level of housing subsidies is determined by macroeconomic conditions rather than policy choices. Moreover, the subsidy rate tends to increase at the "wrong" times, i.e., whenever the inflation rate increases or financial markets tighten. Finally, a large share of the FONAVI and BHN transfers are financed through the operation of indexation schemes that essentially provide implicit insurance programs. These programs protect households from changes in real interest rates and real wages rather than changes in the inflation rate. The reliance on this indirect mechanism to finance mortgage loan forgiveness reduces the credibility of the financial system.

The objective of housing policy reform should be to contain both the direct and prospective effects of housing market functioning on the rest of the economy. This requires that attention be given to fiscal, financial, and subsidy targeting concerns. On the mobilization side, efficient and explicit fiscal instruments, such as direct rather than payroll taxes, should be used to mobilize resources for the subsidies given to this sector. Similarly, well-defined financial mechanisms, such as interest rate risk insurance programs, that mandate that households share a greater part of inflation or real income risks, could substantially reduce the costs to government of providing housing finance. Finally, on the subsidy targeting side, subsidies should be distributed to offset the adverse consequence of severe financial conditions. By targeting much smaller subsidies on those lower-income households who can provide evidence that they are closest to the "margin," it seems very likely that a greater number of the more than one million households who are eligible for FONAVI subsidies can benefit most.

One way to bring those various components of effective policy together would be first, to separate the subsidy distribution from the financial mechanism; and second, to give priority to those below a certain income level who were willing to contribute some of their own resources. For instance, if a 5,000 austral per unit subsidy were sufficient to induce household resource mobilization and production, then the subsidies could induce as many as 50,000 units per year--30,000 more than FONAVI produced in 1985--with less than half of the revenues FONAVI currently collects. If this increased level of production reduced the rate of increase in real rents, it would substantially reduce the welfare losses implied by the currently constrained housing market.

The other half of the revenues FONAVI collects could be used to finance the implicit subsidies associated with mortgage indexation schemes. For example, these FONAVI funds could be used to build an insurance reserve to make payments to lenders in the event that the mortgage repayment index caused the real value of the loan to decline. Lenders, in turn, could pass on any such payments from the insurer to

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30/ Diaz-Alejandro (1985) discusses the inflationary behavior of increasing real wages during a period of declining GDP growth.

depositors in positive real interest rates. This result could help reverse the contraction in the financial system.

As Capozza and Gau (1984) show, the insurer could use program design to encourage lenders to reduce the need to rely on FONAVI funds. For example, households could buy insurance for only very large interest rate increases, and take responsibility for smaller increases. In this case the insurance premium would be much lower than is the current essentially "no deductible" approach. In addition, if the economy recovered and real wages increased, there would be no demands on the insurance funds, and an interest-earning reserve could accumulate. This reserve, rather than implicit taxes on depositors or the implicitly subsidized rediscounts of the Central Bank, would help finance the risks that households would not be able to pay for future mortgage payment increases. The adequacy of such a reserve depends upon both what transpires in the economy, as well as the design and targeting of the mortgage instrument.

In the long-run it would be important to try to structure this fund to be self-sufficient, as was recently proposed in Canada. In the short-run, however, there is an advantage to such a fund even if it is not financially self-sufficient. The advantage is that a larger, unmeasured, and more macroeconomic-sensitive subsidy already exists. By attempting to measure and control this subsidy and its attendant implicit taxes, attention would be paid to a housing finance subsidy that has disruptive effects on the financial system.

VII. CONCLUSION

In inflationary economies such as Argentina finance for a long-term assets such as housing is often indexed or subsidized or both. Policy problems arise because implicit subsidies are usually intertwined with the indexation schemes. Moreover, these subsidies are poorly-targeted, very large, not transparent, or even usually measured by traditional fiscal accounting concepts. Large implicit taxes are used to finance these subsidies while the revenues of relatively more efficient tax mechanisms are distributed in even more poorly-targeted inframarginal subsidies. This method of subsidy distribution can have very high resource costs when other financial policies--such as very high real borrowing costs and negative returns to financial assets--cause the housing market to be in disequilibrium for extended periods of time.

In Argentina the contribution that restructuring of the housing finance system could make to eliminating the imbalances in macroeconomic policy could be very significant. It does not appear to be a coincidence that the financial system is shrinking in size and becoming increasingly expensive at the same that the market for one of the Nation's largest forms of wealth, housing, is experiencing similar trends. Improving the efficiency of the intermediation for this long-term and currently high return investment could help reverse these trends within the sector. It could also provide greater incentives to once again use the financial system.

THE ARGENTINE HOUSING MARKET

The Supply of New Housing. In 1985 the number of housing units produced decreased to the lowest level reached in 25 years. This level of production is almost certainly less than sufficient to keep the housing stock intact much less provide for the housing needs of new households. Between 1976 and 1985 total housing production dropped by 60,000 equivalent units to less than 100,000, and the public sector share of production doubled to 50,000. New housing starts financed by the private sector hit very low levels in 1984-85, and output was well below the estimated household formation of 125,000 per year. For example, between 1970 and 1980, the housing stock increased by one million units and the number of new housing units produced during these years was 1.72 million. The difference between the additions of the stock during the period and the increase in the size of stock at the end of the period, i.e., 720,000 units, implies a depreciation rate of the existing housing stock on the order of 1.2 per annum. The "equivalent units" produced in 1985 barely reach this figure, and the actual number of units produced was well below this figure. A similar depreciation rate is used in Castaneda and Quiroz (1986) for Chile.

The growth in the number of households relative to the growth in the stock of housing suggests that even with much lower headship rates and less space demanded per household it is likely that there was a relative increase in the demand for housing units. In addition, while the real cost of capital generally increased, it is also the case that the relative return to housing investments increased significantly. As is discussed in the text, the negative rates of return on almost all other domestic forms of saving made housing one of the few domestic assets whose return was not taxed by the interaction of inflation with nominal interest rate ceilings.

The Supply of Existing Housing. In 1985 the 104,000 apartments offered for rent in Buenos Aires was equal to 55 percent of the number available in 1981, following reductions in the number of rentals of over 80,000 and 50,000 in 1982 and 1983, respectively. It is interesting to compare these reductions in the number of rental units available in Buenos Aires with total housing production in the country. The reduction in the capital city between 1982 and 1983, for example, was equal to 80 percent of the average annual production for the country for the 1982-1985 period. It was also larger than the average level of national private production during this period. Hence, in one year, four out of every five new housing units added to the housing stock in the entire country were offset by a withdrawal from the stock in the capital. The reduction in the number of units available over 1982-83 was equivalent to reducing the housing stock in the capital by more than 5 percent, and in the Nation by 1.5 percent.

Real Rent Trends. In 1980, before the contraction in housing production, one third of households were not house owners, and 5 percent of families lived with more than one family to a housing unit. These non owners have faced increasingly expensive housing costs. Between 1980-1985 real rents in Greater Buenos Aires increased by more than 16 percent per annum. In contrast, between 1976-1980, a period during which the binding rent control of the preceding 30 years was phased-out, real rents increased by about 4 percent per annum. Real rents in 1985 were 140 percent higher than in 1980.

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